

Solving systems of equations with a variety of methods



1 The point of intersection of two lines.

2 -3

3 -4

4

a

x	$y = 3x - 1$	$y = -5x + 23$
1	2	18
2	5	13
3	8	8
4	11	3
5	14	-2

b $x = 3, y = 8$

5

a

x	$y = 5x$	$y = 9 - 4x$
1	5	5
3	15	-3
5	25	-11

b $x = 1, y = 5$

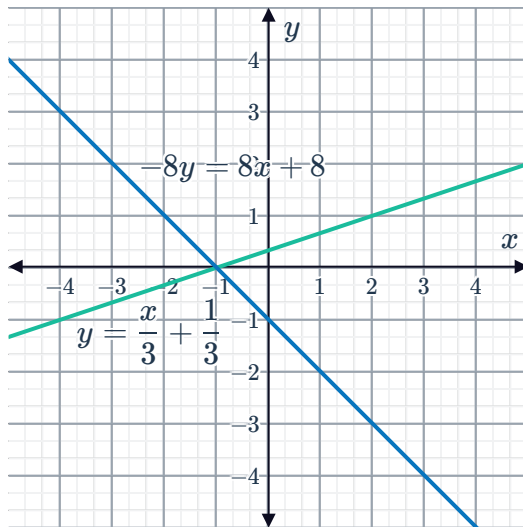
6

a $(-1, 0), \left(0, \frac{1}{3}\right)$



b $(0, -1), (-1, 0)$

c



d $x = -1, y = 0$

7

a $x = 6, y = 4$

b $x = 0, y = -2$

c $x = 4, y = 0$

8

a $x = 8, y = 3$

b $x = \frac{20}{27}, y = \frac{62}{27}$

c $x = -20, y = 15$

d $x = 2, y = \frac{5}{3}$

e $p = \frac{2}{5}, q = \frac{3}{4}$

f $p = \frac{3}{5}, q = \frac{4}{5}$

9

a $x = 4, y = 2$

b Yes

c Yes

10 $x + y = 9, y - x = 1$

11

a $x = 13$

b $x = 9$

12

a 4

b $x = -3, y = -9$

c No

13



a $x = 1, y = 9$

b $y = -3x + 12$

c $3y = 3x + 24$

d $x = 1, y = 9$

e Yes, Equations 3 and 4 are the result of applying identical operations to both the left hand side and the right hand side of each of the original equations.

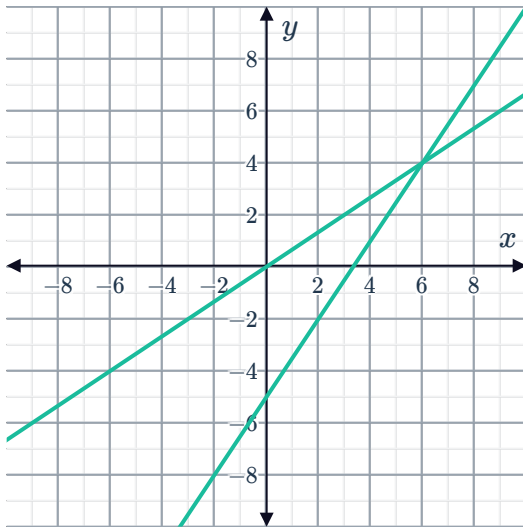
14

a $x = 6$

b Equation 1 and Equation 4

c

d $x = 6$



15

a

b

i

i

No

ii

No

iii

Yes

iv

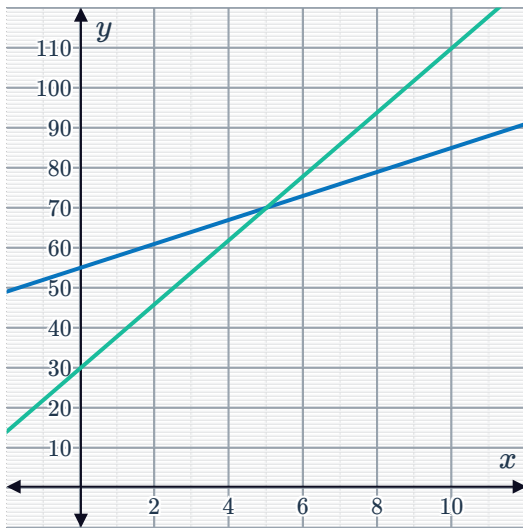
No

x	1	3	5	7	9
y	58	64	70	76	82

ii

x	1	3	5	7	9
y	38	54	70	86	102

c



5 minutes

e TeleNow

16

a No

b Yes

c No

d No

Additional questions

17

a (0, 4)

b (4, 6)

c (8, 9)

d (-3, 4)

e (7, 2)

f (9, 6)

18

a (5, -1)

b (-2, 6)

c No solutions

d (7, -6)

19

a (-3.125, 7.9375)

b $\left(33, -\frac{100}{19}\right)$

c $\left(\frac{13}{3}, 36\right)$

d $\left(\frac{15147}{37}, \frac{347}{37}\right)$